

Abdelrahman Abozena

AI & Data Science | LLM Agents, RAG, Machine Learning & Python/FastAPI Systems

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SUMMARY

AI-focused Computer Science student and software engineer with practical experience building LLM agents, RAG workflows, multi-agent simulations, machine learning pipelines, data analysis projects, and scalable backend APIs. Project-based portfolio covering Python, EDA, ML evaluation, deep learning foundations, NLP/Transformers, Generative AI, and deployment with FastAPI/Streamlit.

TECHNICAL SKILLS

AI / LLMs	Generative AI, LLM Agents, RAG, LangGraph, Prompt Engineering, Multi-Agent Systems, Local LLMs/Ollama, Transformers, NLP
ML / Data Science	Machine Learning, Deep Learning, EDA, Data Preprocessing, Feature Engineering, Model Evaluation, CTR Prediction, Sentiment Analysis, Computer Vision/YOLO
Backend / MLOps	Python, FastAPI, Node.js, Express, REST APIs, WebSockets, Kafka, Streamlit, real-time inference workflows, backend system architecture
Databases / Tools	MySQL, MongoDB, SQLite, Git, GitHub, Docker basics, React, Vite, TypeScript, Tailwind CSS

AI / DATA SCIENCE TRAINING

Harbin Institute of Technology | Generative AI Training Program / AI Summer School | On-site in China

Focused on Generative AI, LLMs, probabilistic models, and applied AI concepts through in-person training.

GCI | AI & Data Science Course - Japanese Online Program | Present

Currently studying AI and data science through a Japanese online program.

Digital Egypt Pioneers Initiative (DEPI) | AI / Machine Learning Track | Nov 2025 - Present

Applied track covering ML, deep learning, NLP, computer vision, MLOps concepts, and project-based assignments.

Instant Software Solutions | AI & Data Science Training Program | Jan 2026 - Present

Training across data science, machine learning, NLP, Transformers, LLM automation, and backend AI systems.

Orange | Data Science Training Course | Sep 2024 | Egypt

Completed data science training in a professional telecom environment.

SELECTED AI / ML / DATA SCIENCE PROJECTS

OrchCode Studio - Multi-Agent Coding System

TypeScript, Rust, LLM Agents, Local LLM Providers, Agent Orchestration

- Built a local-first multi-agent coding environment that turns ideas into code through structured orchestration, task planning, safe execution, and provider-agnostic LLM workflows.
- Designed agent flows for decomposition, planning, coding, validation, memory/context handling, and iterative improvement to help lighter or local models produce stronger coding outputs.
- Focused on practical coding-agent reliability: controlled execution, organized workspaces, agent memory, protocol packages, scripts, docs, and test outputs.

AgenticAI Community Simulator

Python, FastAPI, React, Vite, WebSockets, MySQL, Multi-Agent Systems

- Built a full-stack multi-agent simulation platform that creates AI personas/communities to test ideas, market reactions, and social opinion evolution.
- Agents can discuss, object, support, influence each other, and produce qualitative reasoning instead of only returning a static score.
- Includes real-time WebSocket simulation events, research context, Idea Court, Persona Lab, dashboards, authentication, credits/admin controls, and analytics-style UI.

AdClick Prediction System / Project Fusion ECU

Python, Jupyter, Data Fusion, Scikit-learn, XGBoost, Streamlit

- Built a leakage-safe, time-aware CTR prediction pipeline using fused user profiles, ad metadata, and historical behavior data.
- Covered the full data science workflow: cleaning, EDA, preprocessing, feature engineering, target encoding, model training, evaluation, and deployment.
- Delivered a Streamlit decision console for scoring CTR, ranking Top-K ad spend, explaining individual predictions, and exporting selected/skipped records.

Real-Time Comment Analysis & Fake Comment Detection Pipeline

FastAPI, Kafka, Python Workers, Ollama, SQLite, React/Vite

- Built an end-to-end analytics pipeline for comment sentiment, suspicious/coordinated behavior detection, alerts, and live monitoring.
- Used Kafka topics, background Python workers, FastAPI dashboard APIs, local Ollama integration, SQLite state, and a React/Vite monitoring interface.

Ask Your PDF with Transformers

FastAPI, Transformers, PyTorch, PyPDF2, Tailwind UI

- Built a document QA app where users upload PDF/TXT files, ask questions across the full document or selected sections, and inspect supporting context.
- Implemented backend extraction, section detection, transformer-based QA, and a three-column interface for documents, answers, and annotations.

Prompt Optimizer for Coding Agents

Python, Streamlit, LLM Prompting, Coding-Agent Workflows

- Built a tool that analyzes code diffs and user intent, asks clarifying questions, and generates precise final prompts for AI coding agents.
- Useful for teaching prompt engineering, context gathering, and how to turn vague development requests into structured agent instructions.

Robot Vision / Real-Time Object Detection

Python, Computer Vision, Object Detection, Video Analytics

- Built an AI vision project for real-time object detection in robot point-of-view videos, enabling perception, tracking, and scene understanding.
- Demonstrates computer vision fundamentals, detection pipelines, and practical analysis of dynamic environments.

Agentic Router Attack Simulation

Agentic AI Safety, Simulation, Prompt/Task Tampering

- Simulated how router manipulation and task tampering in agentic AI systems can lead to sensitive data leakage.
- Highlights AI safety awareness for multi-agent systems, tool routing, and secure agent workflow design.

PROFESSIONAL EXPERIENCE

EgenSolve | Back End Developer | Mar 2025 - Sep 2025 | Giza, Egypt

- Developed a centralized backend system to manage employee accounts and system issues.
- Linked users directly with developers for faster problem resolution through backend workflows and APIs.

Crossover | React Developer | Oct 2022 - Dec 2023 | Remote

- Developed React-based user interfaces and reusable frontend components in a remote environment.

EDUCATION & CERTIFICATIONS

Egyptian Chinese University | B.S. in Computer Science - AI-focused specialization/coursework

- Google x Kaggle - AI Agents Intensive Certification
- AI Summer School - Generative AI & Probabilistic Models - Harbin Institute of Technology